Unleashing Pretrained Generative Priors for Single-Image HDR Reconstruction via Uncertainty-Routed Attention

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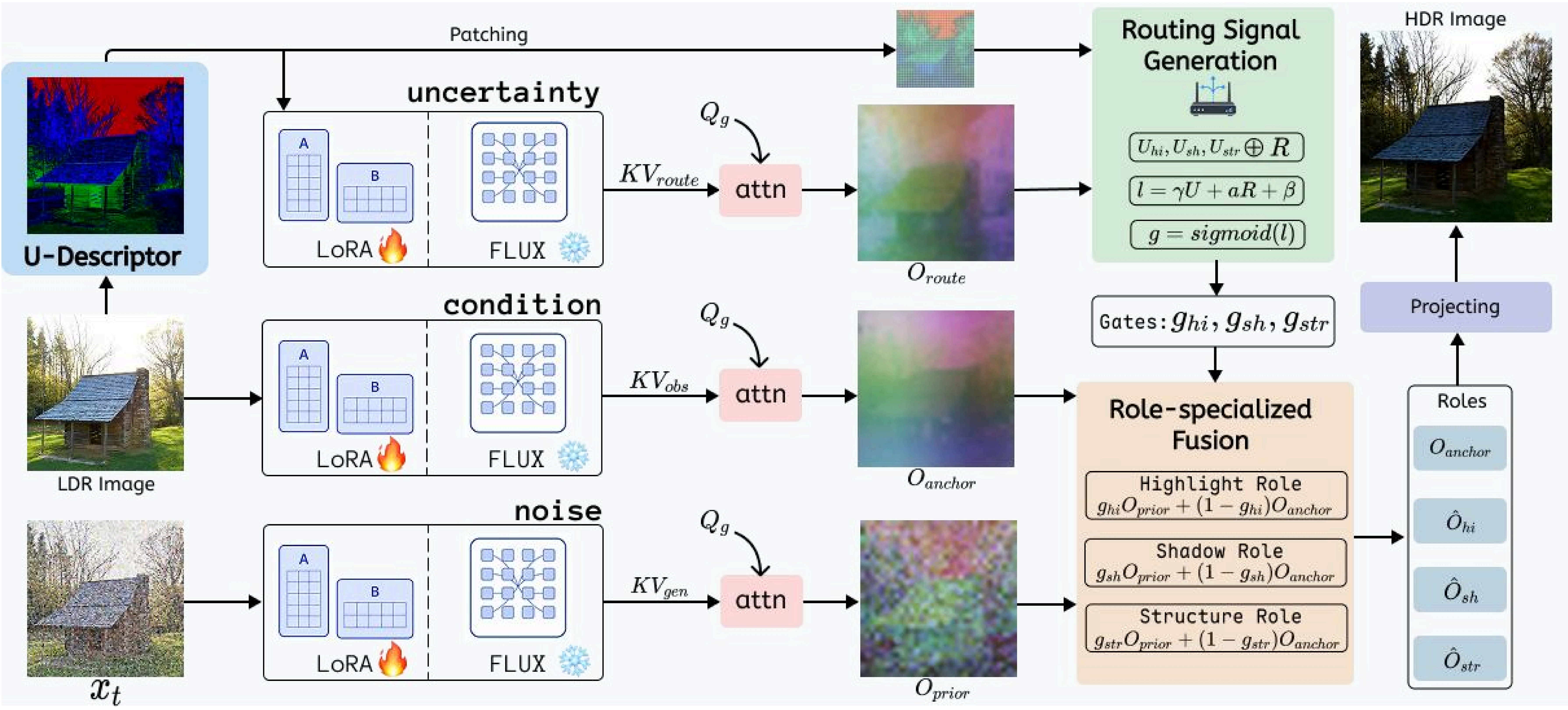
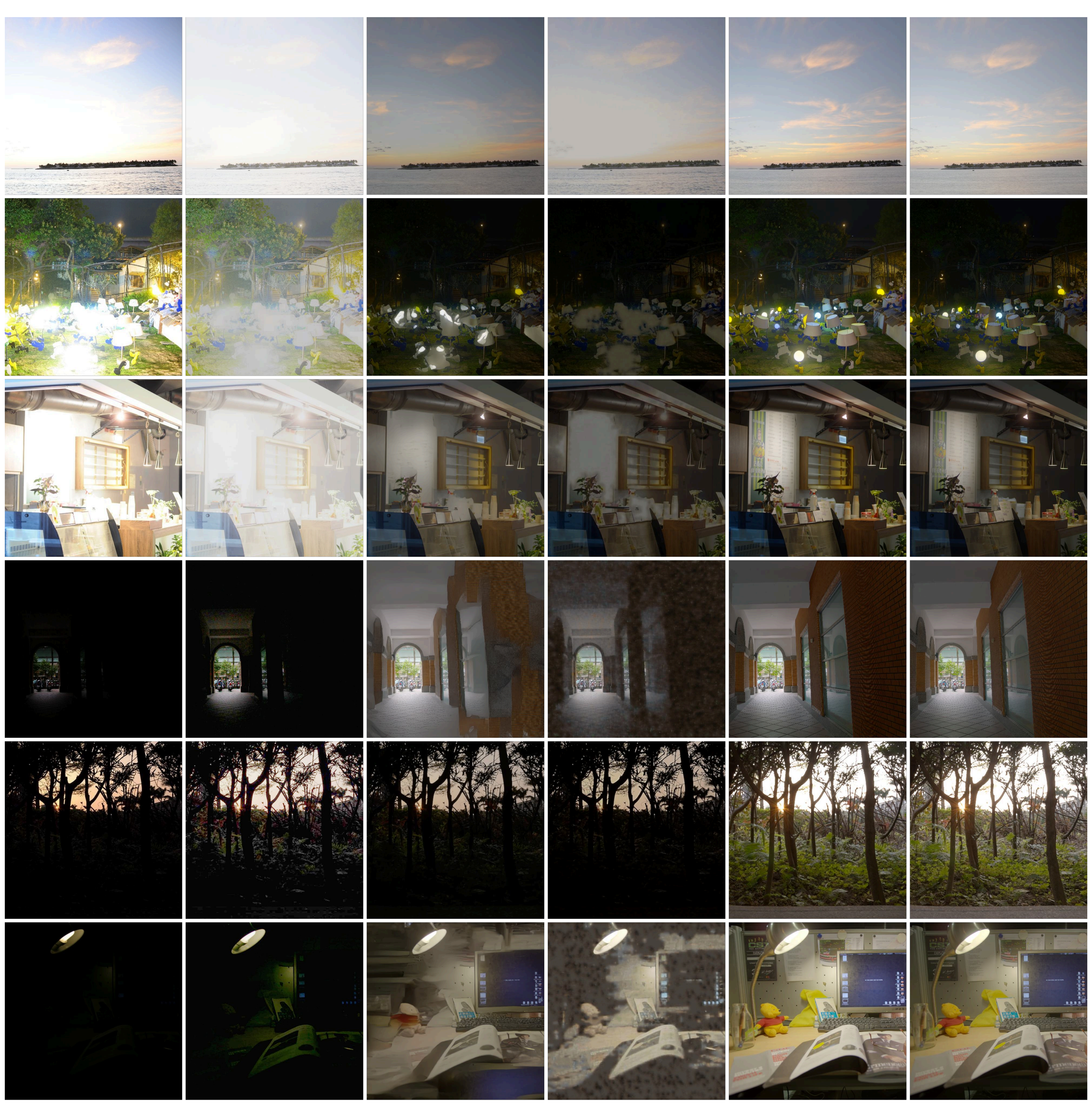
Method



1. Unleashing VLM's pretrained priors
2. Uncertainty-Routed Attention

Qualitative Comparsion



(a) Input (b) HDRCNN (c) DDPM (d) Reti-Diff (e) Ours (f) Reference

Quantitative Comparsion

Dataset	Method	LPIPS↓	MUSIQ↑	NIQE↓	CLIP-IQA↑	MANIQA↑	HDR-VDP-3↑
HDR-Real	HDRCNN	0.3497	54.88	7.04	0.4434	0.4109	5.71
	SingleHDR	0.3156	51.45	6.19	0.3370	0.3356	6.16
	ExpandNet	0.3416	51.67	6.07	0.3706	0.3397	5.91
	HDRUNet	0.3937	55.66	**5.12**	0.2545	0.3446	4.41
	DDPM	0.1921	64.84	5.75	0.4965	0.4915	7.45
	DDIM	0.2365	**67.53**	5.23	0.4643	0.4975	6.76
	HDR-Trans.	0.3665	54.02	5.31	0.3774	0.3451	5.69
	Reti-Diff	0.2645	54.12	5.37	0.3994	0.3644	7.31
	Ours	**0.1097**	63.48	5.27	**0.5811**	**0.5017**	**8.57**
HDR-Eye	HDRCNN	0.2811	66.01	4.53	**0.5669**	**0.5666**	7.08
	SingleHDR	0.2436	60.43	4.21	0.4317	0.4193	7.66
	ExpandNet	0.3105	54.87	5.44	0.3879	0.3061	7.32
	HDRUNet	0.3054	66.60	4.45	0.3299	0.4683	5.79
	DDPM	0.2005	68.54	4.76	0.4904	0.5057	7.99
	DDIM	0.2007	69.00	4.82	0.4731	0.4936	7.92
	HDR-Trans.	0.2537	66.17	**3.73**	0.5011	0.4458	7.51
	Reti-Diff	0.2475	67.55	4.14	0.5257	0.4979	7.74
	Ours	**0.1311**	**70.71**	4.39	0.5398	0.5260	**8.32**
AIM 2025	HDRCNN	0.3605	39.98	8.72	0.3191	0.3229	6.57
	SingleHDR	0.2460	43.53	6.74	0.3748	0.4043	7.90
	ExpandNet	0.2149	44.91	7.08	0.3734	0.4104	8.21
	HDRUNet	0.2218	49.22	7.24	0.3412	0.4348	7.06
	DDPM	0.1286	57.36	8.83	0.5225	0.5601	**8.78**
	DDIM	0.1580	**58.65**	8.61	0.5076	0.5606	8.36
	HDR-Trans.	0.2604	43.67	6.81	0.3867	0.3990	7.78
	Reti-Diff	0.2332	46.25	6.45	0.3662	0.4024	8.19
	Ours	**0.1139**	58.20	**6.12**	**0.5436**	**0.5633**	8.48

Method	LPIPS↓	MUSIQ↑	NIQE↓	CLIP-IQA↑	BRISQUE↓
Zero-DCE	0.3351	55.85	7.76	0.6111	27.3925
PairLIE	0.2434	56.85	4.09	0.3238	20.2068
SCI	0.3392	57.17	7.87	**0.6200**	25.6633
Quadprior	0.2143	63.62	5.49	0.4110	16.8182
AGLLDiff	0.1933	71.24	6.39	0.2853	38.5621
Retinexformer	0.1314	63.14	3.74	0.4262	16.8477
Diff-Retinex	0.0989	71.95	5.03	0.4947	26.5251
Reti-Diff	**0.0958**	71.90	5.21	0.4295	17.7525
Ours	0.1395	**72.71**	**3.62**	0.4778	**12.9454**